

$$(\hat{p}-1) B^{\alpha} (i\ell-1) + (\hat{p}) B^{\alpha} i\ell \sum_{i=1}^{\infty} \frac{1}{i^{\alpha}}$$

Deep Learning

$$L = -[y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

$$\frac{dL}{d\hat{y}} = -\left(\frac{d}{d\hat{y}} y \log(\hat{y}) + \frac{d}{d\hat{y}} (1-y) \log(1-\hat{y})\right)$$

$$= -\left(\frac{y}{\hat{y}} + (1-y) \frac{-1}{1-\hat{y}}\right)$$

$$= -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}}$$

$$\frac{dL}{d\hat{y}} = \frac{\hat{y} - y}{\hat{y}(1-\hat{y})}$$